

A Guess, Then a Measurement: The First-Law Kernel Follows the Region’s Shape

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Abstract

I guessed that we need a shape, not a sphere. That is a guess. Casini–Huerta–Myers is a *ball* theorem. Paper 37’s CHM win was on balls.

Same fixed- H occupation transfer. $N = 12$. Balls of radius 2 still select $R^2 - r^2$ ($R = 0.018$ vs flat 0.102). Cubes of side 3: a product of 1d interval weights beats the exported ball kernel and beats flat (0.022 vs 0.047 vs 0.063). Taxicab $t = 2$: the ℓ_1 parabola is slightly better than the export.

Auditor, $N = 10$: ball control CONFIRMED. Cube export REFUTED. Cube-native CONFIRMED.

The guess is now a fact about *this probe*: the winning weight knows the region’s shape. It is still a guess about gravity. Not $8\pi G$. Not de Sitter. I did not write a cube CHM theorem.

1 What this is

Paper 33 already found that S itself knows the shape beyond cut area. This note asks the first-law question on the same families. Exporting the sphere kernel onto a cube is not robust. A cube-native weight is.

Admission. *The shape remark was my guess. I ran the test that can kill it. It survived as a lattice measurement and I still did not promote it to Einstein.*